

Conservative averaging-reconstruction techniques (Ring Average) for 3-D finite-volume MHD solvers with axis singularity

Binzheng Zhang^{a,b,c,*}, Kareem A. Sorathia^d, John G. Lyon^e,
Viacheslav G. Merkin^d, Michael Wiltberger^c

^a Department of Earth Sciences, the University of Hong Kong, Hong Kong SAR

^b Laboratory for Space Research, the University of Hong Kong, Hong Kong SAR

^c High Altitude Observatory, National Center for Atmospheric Research, United States of America

^d Applied Physics Laboratory, the Johns Hopkins University, United States of America

^e Department of Physics and Astronomy, Dartmouth College, United States of America

ARTICLE INFO

Article history:

Received 1 March 2018

Received in revised form 10 July 2018

Accepted 10 August 2018

Available online 28 August 2018

Keywords:

Finite-volume method

Numerical MHD

Spherical geometry

Axis singularity

ABSTRACT

In cylindrical/spherical geometries, MHD solvers using explicit finite-volume methods face restrictive time steps imposed by the CFL condition due to the clustering of cells in the azimuthal direction near the pole/axis. We use a conservative averaging-reconstruction method (Ring Average) on structured cylindrical/spherical mesh to remove this severe time step restriction for multi-dimensional curvilinear finite-volume MHD solvers. The Ring Average technique is implemented as a post-processing step and thus requires no changes to the existing data structure, grid definition or numerical methods in the original MHD solver. This paper describes the Ring Average algorithm and presents simulation results using field loop advection and MHD blast waves in cylindrical/spherical geometries for validation. The algorithm is shown to be inexpensive and easily implemented in existing curvilinear finite-volume MHD solvers.

© 2018 Elsevier Inc. All rights reserved.

1. Introduction

A large number of MHD flow problems are naturally defined in cylindrical or spherical coordinates such as planetary magnetospheres and rotating magnetic disks. While MHD solvers with Cartesian meshes are used extensively to study these problems, they can be inefficient either due to excessive amount of grid cells required, or can be inconvenient due to difficulties in the treatment of boundary conditions, e.g., fitting a sphere within a Cartesian mesh as an inner boundary. However, a cylindrical or spherical mesh, when appropriate for the problem, still has two major numerical difficulties to overcome: (a) a grid singularity at the axis/pole, and (b) the severe time-step limitation from the clustering of azimuthal cells near the axis/pole.

To address issue (a) using finite-difference methods, special treatments for polar/cylindrical singularities have been developed [1–3]. Another way to address issue (a) is to use finite-volume schemes to solve the MHD equations such that cell

* Corresponding author at: Department of Earth Sciences, the University of Hong Kong, Hong Kong SAR.

E-mail address: binzh@hku.hk (B. Zhang).

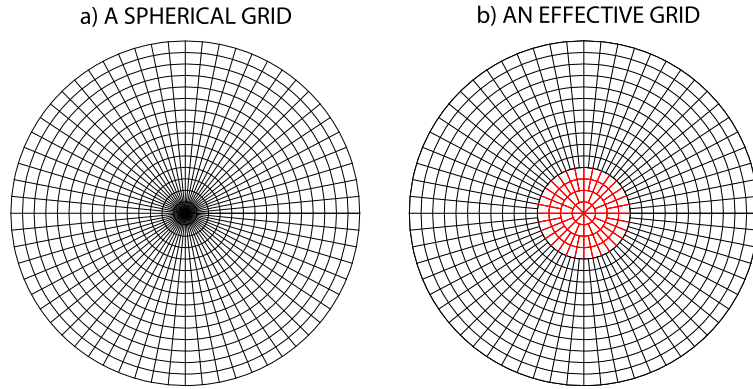


Fig. 1. a) An original spherical polar grid and b) the effective polar grid.

centers are not located on the axis. With finite-volume MHD discretization, the axis singularity of the equations in cylindrical/spherical geometry is easily resolved by setting fluxes from the degenerated faces at the axis to zero. On the other hand, the issue (b) of restrictive time-steps, however, still needs to be addressed. For hydrodynamic simulations, the time step restriction problem can be resolved, for example, using local cell-merging methods [4], or conservative Fourier filtering techniques [5] in order to remove large wave number components in the azimuthal direction near the axis, which may be extended to MHD problems. However, in MHD simulations additional difficulties occur when the Maxwell's equations are involved, since $\nabla \cdot \mathbf{B} = 0$ is strictly required at each time step.

In this note, we developed a post-processing type of conservative averaging-reconstruction algorithm, which is named as the “Ring Average” technique, to address the time-step limitation for 3-D MHD solvers in cylindrical or spherical geometry without changing the existing data structure, grid or numerical schemes of the MHD solver. The Ring Average technique was originally developed by Lyon et al. [6] in the Lyon–Fedder–Mobarry (LFM) global MHD code for magnetosphere simulations, which solves the clustering of “thin” azimuthal cells problem by first averaging conservative quantities within these “thin cells” into “larger cells” in the azimuthal direction and then reconstructing monotonically back to the “thin cells” within each “larger cell”. The original “Ring Average” technique was implemented in the LFM code using a piecewise constant reconstruction in the early 1990s. In this note we extend the original Ring Average technique up to 4th-order accuracy, which is suitable for simulating more general MHD flows with larger parameter space. The Ring Average technique is conservative, computationally inexpensive and can be easily implemented in an existing finite-volume, curvilinear MHD codes without changing the original solver. The Ring Average technique also works well with constrained transport to maintain zero divergence of the magnetic field ($\nabla \cdot \mathbf{B} = 0$), which can also be applied to MHD solvers with other divergence cleaning methods. This note is organized as follows: Section 2 describes the numerical method and gives the formulation of the averaging-reconstruction procedure. Section 3 shows test results of the algorithm using the standard magnetic field loop advection test in cylindrical geometry and the MHD blast wave test in both cylindrical and spherical geometry. Section 4 concludes the note.

2. Method

A polar view of one 3-D shell of a spherical finite-volume grid with 64 cells (controlled volumes) in the azimuthal direction is shown in Fig. 1a. It is clear that in Fig. 1a the clustering of cells in the azimuthal direction near the axis makes the effective length of the corresponding controlled volumes very small. In the example grid shown in Fig. 1a, the ratio between the average cell lengths (defined as volume divided by mean face area) at the equator and near the pole is approximately 100:1, resulting in the explicit time step given by the Courant–Friedrichs–Lewy (CFL) condition being restricted by the inner most cells near the pole axis. In a 3-D simulation with $N \times N \times N$ (radial $\times$ polar $\times$ azimuthal) cells, the time step Δt is proportional to $1/N^2$ rather than $1/N$, assuming the time step is not determined by the radial direction. Thus doubling the grid resolution results in a factor of four decrease in the time step. The Ring Average technique mitigates this effect by redistributing the conserved quantities within a limited range in the azimuthal direction near the axis. An “effective” grid after applying the Ring Average technique in the first four azimuthal rings is shown in Fig. 1b. In this example grid, the cell next to the axis is averaged over octants, the next one is averaged over 1/16-th of the azimuthal range, and so on. Therefore the time step of the simulation is no longer restricted by the clustering of the azimuthal cells right next to the axis, rather, it is determined by the mean cell length of the fifth azimuthal ring, since small scale variations are smoothed out within each effective azimuthal ring where Ring Average is applied. Note that the actual computational grid used in the 3-D finite volume solver is still the standard spherical grid as shown in Fig. 1a. The Ring Average technique provides a post-processing type of treatment to the MHD variables near the axis after the full finite-volume update such that the effective grid follows the example shown in Fig. 1b.

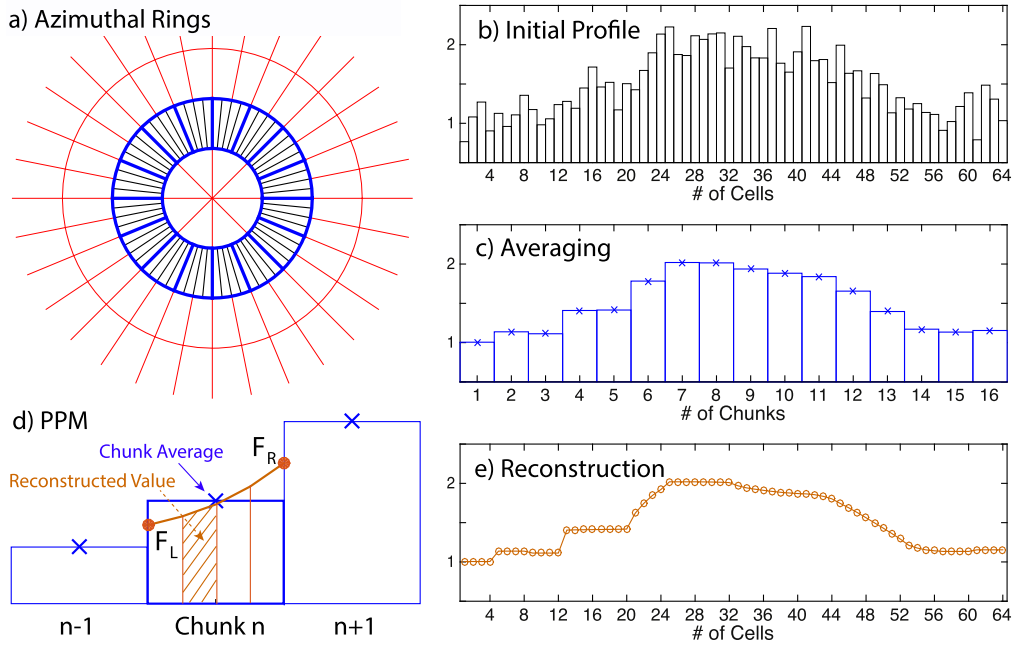


Fig. 2. The Ring average procedure in the azimuthal direction. Panels b, c, e can be regenerated using the Matlab code *ringav_example.m* in the supplemental material.

2.1. Averaging fluid variables

In finite-volume MHD solvers, the volume-averaged conservative fluid variables (mass, momentum, energy) are usually defined at cell centers. Thus the averaging-reconstruction of the fluid variables is essentially a one-dimensional process along the azimuthal direction. After the finite-volume solver update, the conserved fluid variables in each ring are averaged within effective cells (or chunks) and then reconstructed monotonically. The reconstruction method of the conserved quantities can be piecewise constant method (PCM), piecewise linear method (PLM) or piecewise parabolic method (PPM) [7]. In this note, we choose the PPM reconstruction which provides high-order accuracy (up to 4th order) in the reconstruction process. Fig. 2 shows the procedure of the Ring Average technique for fluid variables in one selected azimuthal ring as an example. Fig. 2a shows the spatial distribution of the actual computational cells (here we used 64 azimuthal cells as an example) and the averaged chunks (16 chunks) at the second azimuthal ring. Fig. 2b shows the distribution of the initial data (for example density) along the azimuthal direction. Fig. 2c shows the averaged values of the initial data within each chunk. After the averaging step, the parabola within each chunk is computed using the PPM method by finding the left (F_L) and right state (F_R) of the conserved variables at each chunk interface, constrained by the fact that the integrated conserved quantities remain unchanged, which is shown in Fig. 2d. The corresponding values of the actual computational cells within each chunk are then calculated by integrating the parabola function analytically in a piecewise way, as shown in Fig. 2e. It is clear that after the averaging-reconstruction steps, small variations within each chunk are removed while the total amount of conserved quantities remain unmodified. Note that a monotonic reconstruction method is required to keep the reconstructed solution non-oscillatory within each chunk. For example if the initial profile is a step function, the PPM reconstruction guarantees that the reconstructed cell profile follows exactly the step function.

Using the ring average technique illustrated in Fig. 2, the time step is recomputed based on the pre-defined effective polar grid as shown in Fig. 1b, which can be orders of magnitude larger than the original time step (see Section 1). Assuming the fluid variables are defined at cell centers, the algorithm of reconstructing conserved fluid variables are summarized as follows:

1. Pull data into one-dimensional arrays from the first azimuthal ring right next to the axis (Fig. 2b);
2. Average conserved variables (density ρ as an example) within each chunk to get ρ_{Mean} (Fig. 2c);
3. Apply the standard PPM reconstruction procedure [e.g., [7]] to the averaged values ρ_{Mean} and find interface values ρ_L and ρ_R in each chunk. Together with the conservation constraint, a unique parabola function ($Ax^2 + Bx + C$) that connects ρ_L and ρ_R is determined (Fig. 2d);
4. Integrate the parabola function calculated in Step 3 piece-wisely, reconstruct the conserved variables in each chunk using the following function (Fig. 2e):

$$\rho^r(k_l) = \frac{A}{3N_C^2} (3k_l^2 - 3k_l + 1) + \frac{B}{2N_C} (2k_l - 1) + C \quad (1)$$

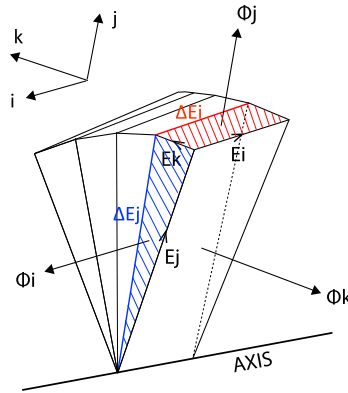


Fig. 3. An example finite-volume chunk for Ring Average with staggered grid.

where $\rho^r(k_l)$ is the reconstructed conserved density in the k_l -th cell within each chunk, with k_l the local index ranging from 1 to N_C , N_C is the number of cells within each chunk. A , B and C are the three constants defining the parabola function that connects ρ_L and ρ_R within each chunk:

$$A = 3(\rho_L - \rho_R - 2\rho_{Mean}), \quad (2)$$

$$B = 2(3\rho_{Mean} - \rho_R - 2\rho_L), \quad (3)$$

$$C = \rho_L; \quad (4)$$

5. Go to the next ring and repeat Step 2–4 until ring average is not needed.

Here ρ_{Mean} is the average density within each chunk. It is relatively straightforward to show that within each chunk, $\sum_{k_l=1}^{N_C} \rho^r(k_l) = N_C \rho_{Mean}$, which is a consequence of the conservative nature of the Ring Average technique. Note that if a piecewise constant reconstruction method (PCM) is used to redistribute the fluid variables, i.e. using $\rho^r(k_l) = \rho_{Mean}$, the Ring Average technique works as well. The PCM technique is similar to the cell-merging algorithm developed by Asaithambi and Mahesh [4], except it only “merges” cell in the azimuthal direction. Note that the above algorithm assumes constant azimuthal spacing with N_C being an even number. It is also possible to adapt the algorithm to non-uniformly spaced azimuthal coordinate systems with N_C being an odd number.

In a finite-volume solver, the Ring Average procedure described above is applied to the conserved quantities: mass, momentum and energy. The primitive variables such as flow velocity and pressure are then computed from the reconstructed conserved variables. The Ring average procedure can also be applied to the thermal energy instead of the plasma energy without violating the conservation constraint of the equations. In such cases, the total energy is conserved in the “Effective Grid” as shown in Fig. 1b rather than the original grid.

If the magnetic fields are also defined at cell centers, e.g., a divergence cleaning algorithm is necessary to maintain the divergence-free ($\nabla \cdot \mathbf{B} = 0$) constraint, then the magnetic fields may be treated the same way as the fluid variables (except that a divergence clean step is necessary after the Ring Average process). However, for finite volume MHD solvers using staggered grids with magnetic flux/fields defined at cell interfaces, the Ring Average process is more complicated since the divergence-free condition is an additional constraint besides the conservation of magnetic flux, which is described in the next section.

2.2. Averaging magnetic fluxes

When the magnetic fluxes/fields are defined at cell interfaces and the electric fields are defined at cell edges using staggered grids, the magnetic fluxes/fields go through a similar process of averaging-reconstruction as the fluid variables described in the previous section. However, in MHD simulations it is extremely important to maintain $\nabla \cdot \mathbf{B} = 0$ for the magnetic fields. Therefore the averaging-reconstruction of magnetic fluxes/fields requires additional steps when applying the Ring Average technique in order to maintain $\nabla \cdot \mathbf{B} = 0$. In the following sections, we use magnetic flux as an example for illustrating the algorithm since the conservation is more natural in terms of magnetic flux rather than fields. Fig. 3 shows the definition of magnetic fluxes and electric fields in a spherical grid within one averaging chunk near the axis that contains four original “thin” cells as an example. It is straightforward to adapt the following algorithms to finite-volume MHD solvers that compute magnetic fields instead of magnetic fluxes.

As shown in Fig. 3, in a general curvilinear geometry defined in the $(\hat{i}, \hat{j}, \hat{k})$ coordinate system, the i -direction is along the axis and the k -direction is the azimuthal direction where the Ring Average technique is applied. If the Ring Average technique described in the previous section is applied to reconstruct all three components of the magnetic flux (Φ_i, Φ_j, Φ_k),

the divergence free condition is very likely to be violated locally since there is no guarantee that $\nabla \cdot \mathbf{B}$ in each “thin” cell is zero after reconstructing all three components of the magnetic flux separately – even though the total $\nabla \cdot \mathbf{B}$ within each chunk remains zero due to the conservative nature of the reconstruction method. Thus in order to incorporate the $\nabla \cdot \mathbf{B} = 0$ constraint, the conservative averaging-reconstruction procedure is only applied to the i -component (Φ_i) and j -component (Φ_j) of the magnetic flux as shown in Fig. 3. The k -component (Φ_k) of the magnetic flux is then computed through Faraday’s Law using “perturbation” electric fields (ΔE_j and ΔE_i) which correspond to the changes in Φ_i and Φ_j after the reconstruction step.

The algorithm of getting the reconstructed magnetic flux Φ_i^r is similar to the fluid variables, except that an additional step for a ΔE_j component along j -edges is needed to maintain $\nabla \cdot \mathbf{B} = 0$, as shown in Fig. 3. The algorithm is summarized as follows:

1. Start from the first Φ_i ring ($j = 1$) next to the axis;
2. Average Φ_i within each chunk in the k -direction;
3. Use the same PPM reconstruction subroutine in each chunk to compute $\Phi_i^r(k_l)$;
4. Compute perturbation electric field components ΔE_j corresponding to the accumulative change in Φ_i according to Faraday’s Law;

$$\Delta E_j(k_l + 1) = \sum_{k_l=1}^{N_C-1} \frac{\Phi_i^r(k_l) - \Phi_i(k_l)}{\Delta t}. \quad (5)$$

The assumption in Eqn. (5) is that $\Delta E_k \equiv 0$ so the change from Φ_i to Φ_i^r only comes from ΔE_j ;

5. Go to the next ring and repeat 2–4 to get Φ_i^r and ΔE_j until ring average is not needed.

Steps 1–3 are essentially the same as the fluid part of the Ring Average process described in the previous section. Therefore the same reconstruction subroutines as described in Section 2.1 can be directly used. Here Δt is a numerical time scale that corresponds to the change of magnetic fluxes in the reconstruction step which does not necessarily associate with physical time scales of the simulated MHD flows. In practical simulations Δt is simply set to be 1. Note that on the edges of each chunk where $k_l = 1$ and $k_l = N_C + 1$, $\Delta E_j \equiv 0$ due to no changes in the total flux after the conservative reconstruction of Φ_i . Similarly, the algorithms for getting the reconstructed magnetic flux Φ_j^r on j -faces and the ΔE_i component along i -edges as shown in Fig. 3 are summarized as follows:

1. Start from the first non-degenerated Φ_j ring (e.g., $j = 2$) next to the axis;
2. Average Φ_j within each chunk in the k -direction;
3. Use the same PPM reconstruction subroutine in each chunk to compute $\Phi_j^r(k)$;
4. Compute a perturbation electric field components ΔE_i corresponding to the accumulative change of Φ_j according to Faraday’s Law

$$\Delta E_i(k_l + 1) = - \sum_{k_l=1}^{N_C-1} \frac{\Phi_j^r(k_l) - \Phi_j(k_l)}{\Delta t}, \quad (6)$$

5. Go to the next ring and repeat 2–4 to get Φ_j^r and ΔE_i until ring average is not needed.

Again Steps 1–3 are the same as the Ring Average process applied to Φ_i , except that the starting index for Φ_j is $j = 2$ because the $j = 1$ faces are degenerated with zero face areas as shown in Fig. 3. Note that on the edges of each chunk, $\Delta E_i(1) = \Delta E_i(N_C + 1) \equiv 0$ due to no changes in the total flux after the conservative reconstruction of Φ_j . After the reconstruction of Φ_i and Φ_j in all the azimuthal rings that need Ring Average, the changes in the Φ_k component of the magnetic flux is updated using the perturbation electric fields ΔE_i and ΔE_j computed in the above steps through Faraday’s Law.

Note that in the above algorithm of calculating the perturbation electric fields, the k -components of the perturbation electric field are chosen to be zero in the reconstruction process, i.e., $\Delta E_k \equiv 0$, because it does not change the Φ_k component of the magnetic flux. Therefore, the perturbation electric fields along the edges of each chunk are exactly zero. As a result, $\Delta \Phi_k \equiv 0$ on the interfaces of each chunk, and the total $\nabla \cdot \mathbf{B}$ remains unchanged (which is zero) after the update of Φ_k . The above steps guarantees that the magnetic divergence is zero both locally and globally after the reconstruction step.

The above algorithm of averaging-reconstructing magnetic fluxes are applicable to both cylindrical and spherical geometries. For 2-D solvers in cylindrical grids (e.g., only the j - and k -direction are evolved and magnetic fields are in the $j - k$ plane), the only magnetic flux component that needs Ring Average is the Φ_j component. For 2.5-D and 3-D simulations in cylindrical grids, both Φ_i and Φ_j components need the Ring Average technique.

Table 1

An example of Ring Average setup used in the LFM code.

	$m = 1$	2	3	4	5	6	7	8
$N = 32$	$N_C = 8$	16						
$N = 64$	$N_C = 8$	16	32	32				
$N = 128$	$N_C = 8$	16	32	32	64	64	64	64

2.3. Remarks

Choosing the number of averaging chunks As long as the fluid and magnetic variables are conserved using the Ring Average method, the choice for the number of averaging chunks within each azimuthal ring is usually non-unique. Thus the configuration of the averaging chunk is both problem and resolution dependent. One simple rule for determining the degree of averaging near the pole/axis is to have the effective aspect ratio between radial and azimuthal directions close to one. For uniformly spaced cylindrical/spherical geometry (i.e., $\Delta R = \text{const}$, $\Delta\theta = \text{const}$), an effective length for CFL condition calculations, maybe defined as the ratio of the volume of a cell to the area of a face. For a polar grid, with N cells in the azimuthal direction (usually N is chosen to be power of two, i.e., $N = 2^n$), the effective azimuthal length at cell center is calculated as

$$\Delta L_{eff} = \frac{2m-1}{2} \frac{2\pi}{N} \Delta R, \quad (7)$$

where ΔR is the radial spacing and m is the radial cell index with the first cell at $m = 1$. If the first cell is averaged into octants, the effective length becomes $\frac{\pi}{4} \Delta R$, which is close to the radial spacing. The cells going outward have effective lengths in the ratio of 1,3,5,... The strategy for picking the number of cells over which to average is to keep the following ratio:

$$\frac{2m-1}{N} 2^k \pi \approx 1 \quad (8)$$

where 2^k is the number of cells to average over for radial cell number, m . Table 1 lists the number of chunks in each radial cell number m used in the LFM code for different azimuthal cell number N . Note that it is also possible to extend the ring average technique to be applicable to non-uniformly spaced spherical coordinates. In those cases, the configuration of the averaging rings are more complicated but the basic idea remains the same.

Calculation of the time step When applying the Ring Average technique, the time step for explicit finite-volume MHD solvers needs to be recalculated based on the effective grid. If the number of averaging chunks are chosen such that the aspect ratio between the radial and azimuthal direction is always greater than one, i.e., $\frac{R}{\Delta R \Delta\theta_{eff}} > 1$ where $\Delta\theta_{eff}$ is the azimuthal extension of each averaging chunk, the adjusted time step is simply summed in each chunk based on the number of azimuthal cells within each chunk. If the number of averaging chunks does not guarantee that the aspect ratio between the radial and azimuthal direction is always greater than one, the time step needs to be adjusted using the Courant condition, which can be calculated as follows:

$$\Delta t = \min \left(\frac{\Delta L_{eff}}{V_{max}} \right), \quad (9)$$

where V_{max} is the maximum speed within a controlled volume, and ΔL_{eff} is the mean cell length based on the effective grid, which is calculated as:

$$\Delta L_{eff} = \frac{\Delta V}{\min(A_i^*, A_j^*, A_k)}, \quad (10)$$

with ΔV defined as the volume of a computational cell, and A_i^* , A_j^* are the effective face areas in the i , j direction and A_k is the cell face area in the k direction as shown in Fig. 3. Of course it is possible to re-calculate the time step using the CFL condition based on other definitions of a controlled volume. The key idea is to adjust the time step based on the effective grid size after the Ring Average process, which can be orders of magnitude greater than the original time step.

Boundary conditions across the pole/axis The boundary conditions across the pole/axis in cylindrical/spherical geometry are not different from the standard boundary conditions for finite-volume MHD solvers. Therefore no special attention is needed in the boundary condition module when applying the Ring Average method, except that boundary condition routines are needed after the Ring Average procedure so that the information is updated in the boundary cells. Here we only illustrate the boundary conditions across a cylindrical geometry as a reference for implementing the Ring Average method when using staggered grids.

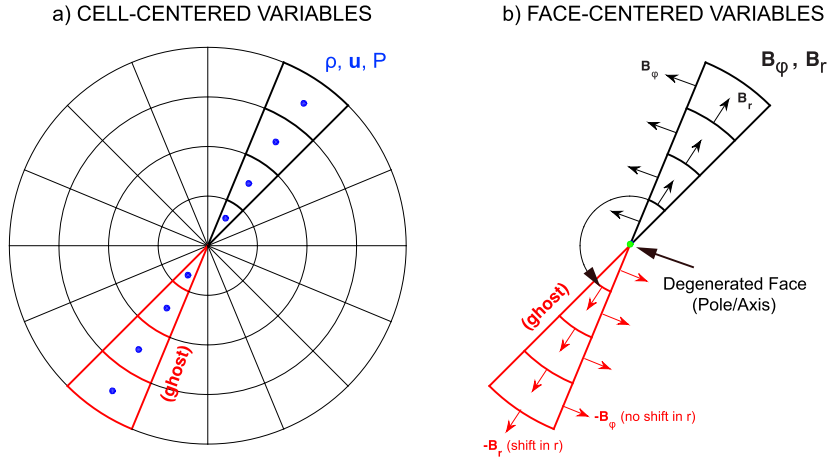


Fig. 4. Boundary conditions across the pole/axis for a) fluid variables and b) magnetic variables. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Fig. 4 shows an example of specifying boundary conditions in the LFM code for fluid and magnetic variables in a 2-D cylindrical coordinates with $N \times N$ cells. The four ghost cells across the pole are shown in red and the corresponding active cells are showing using thick black lines. The cell-centered fluid variables (ρ , $\mathbf{u}$ and P) are shown using blue circles in Fig. 4a. In the LFM code, the finite-volume MHD solver tracks the Cartesian components of the flow velocity (v_x , v_y) and the face-normal components of the magnetic flux (B_r , B_ϕ). Thus all the fluid variables in the ghost cells across the pole are simply taken from the corresponding active cells after 180° rotation around the pole. Assuming the index of active computational cell starts from one, that is, the corresponding index of the four ghost cells shown in Fig. 4 ranges from $-N_g$ to 0, where N_g is the order of the reconstruction divided by two, the density in the ghost cells are specified as

$$\rho(1-n, j) = \rho(n, \text{mod}(j + N/2, N)), n = 1, 2, \dots, N_g \quad (11)$$

Here j is the azimuthal index which ranges from 1 to N . The Cartesian component velocity and pressure in the ghost cells are specified in the same way as density.

Fig. 4b shows the boundary condition specification for the magnetic variables. Here B_r is the normal component in the $\hat{r}$ (radial) direction and B_ϕ is the normal component in the $\hat{\phi}$ (azimuthal) direction in a cylindrical coordinate. Both components in the ghost cells are filled using the same 180° rotating as fluid variables. One important note for specifying magnetic variables is that the sign of the magnetic variables taken from the corresponding active region should be flipped since both the $\hat{r}$ and $\hat{\phi}$ directions change signs across the pole. If a vector is solved in the base Cartesian coordinate system rather than the Cylindrical coordinate system, then the boundary conditions for vector variables are treated the same way as scalar variables. Another note is that when using staggered finite volume discretization, the first normal field (or flux) in the $\hat{r}$ direction is located at the pole, which is degenerate. Therefore the B_r component in the ghost cells are taken from the corresponding active region with a shifted index in i , as shown by the red arrows in Fig. 4b. The same index shift is not needed for the B_ϕ component.

Electric fields on the axis In a staggered finite volume grid as shown in Fig. 3, the electric field components E_i along the axis are co-located and the E_k components are degenerate due to zero edge length at the axis. Therefore the E_i components along the axis are set to be the average value over the azimuthal index k , and the electric field components E_k on the axis are set to zero:

$$E_i(i, j = 1, k) = \frac{1}{N} \sum_{k=1}^{k=N} E_i(i, j = 1, k), \quad (12)$$

$$E_k(i, j = 1, k) = 0. \quad (13)$$

This treatment of the electric field on the axis guarantees that the magnetic flux through the degenerate cell faces at the axis is zero. Without these settings of the electric field components on the axis, some MHD solvers with very low numerical diffusion can be numerically unstable in cylindrical/spherical geometry due to exponential growth of numerical noise generated at the axis, especially in very low- β plasma simulations.

Field loop damping on the axis One other potential problem with a spherical axis is that in some 3-D MHD simulations where the flow stagnates at the axis, there is a tendency for magnetic loops to form around the axis, which can grow to quite large magnitude if left unattended, especially in very low- β plasma situations such as in the inner magnetosphere in

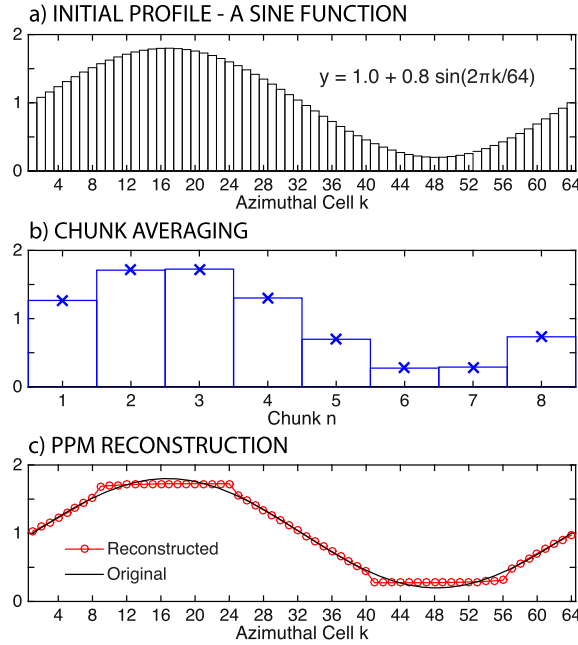


Fig. 5. The clipping of the PPM reconstruction on un-filtered initial profiles.

planetary magnetosphere simulations. One simple method to mitigate the loop formation is to apply a resistive electric field only on the axis after the reconstruction of the Φ_k magnetic flux component in order to enforce that the sum of the Φ_k values around the axis (magnetic field loops) is reduced within several steps:

$$\Delta E_i^{axis} = -A_{fix} \cdot \frac{1}{N} \sum_{k=1}^{k=N} \Phi_k(k) \quad (14)$$

Where A_{fix} is a constant between 0 and 1. In the LFM code for magnetospheric simulations, it is usually chosen to be 0.5 [6]. When A_{fix} is chosen to be 1, the inner most ring of the reconstructed Φ_k component is subtracted by the mean value of the loop fields such that the sum of the Φ_k component is zero. In other words, when $A_{fix} = 1$, any field loop generated at the axis is immediately eliminated after the reconstruction of Φ_k .

Fourier reductions for non-Cartesian vectors When magnetic fields are constant across the axis/pole, the normal components of the magnetic flux Φ_i and Φ_j vary as $\sin(\frac{2\pi}{N}k)$ or $\cos(\frac{2\pi}{N}k)$ functions along the k -direction. Thus applying the Ring Average technique to the sine or cosine functions results in distorted magnetic field directions due to the clipping nature of the monotonic reconstruction algorithm such as the PPM (PCM and PLM are even worse), as shown in Fig. 5. Of course non-clipping limiters seem to be a natural choice to improve the field calculations near the axis. However, a more robust method is to remove the $\sin(\frac{2\pi}{N}k)$ and $\cos(\frac{2\pi}{N}k)$ components in $\Phi_{i,j}(k)$ before applying the Ring Average technique to the i and j -component of the magnetic fluxes. Once Ring Average is applied to the residual magnetic fluxes, the $\sin(\frac{2\pi}{N}k)$ and $\cos(\frac{2\pi}{N}k)$ components are then added back to the reconstructed flux/fields. This simple method keeps the direction of the magnetic fields from being distorted by the PPM reconstruction. In general, a Fourier series expansion of $\Phi_{i,j}$ in the k -direction is written as:

$$\Phi_{i,j}(k) = A_0 + A_1 \cos(\frac{2\pi}{N}k) + B_1 \sin(\frac{2\pi}{N}k) + \tilde{\Phi}_{i,j}(k), \quad (15)$$

where k is the cell index along the azimuthal direction ranging from 1 to N , with N the total number of cells in the azimuthal direction, A_0 , A_1 and B_1 are the corresponding coefficients of a Discrete Fourier Transform with wave number $m = 0, 1$:

$$A_m = \frac{1}{N} \sum_{k=1}^{k=N} \Phi_i(k) \cos(m \frac{2\pi k}{N}), \quad (16)$$

$$B_m = \frac{2}{N} \sum_{k=1}^{k=N} \Phi_i(k) \sin(m \frac{2\pi k}{N}). \quad (17)$$

Thus $\tilde{\Phi}_{i,j}(k)$ is the magnetic flux that contains all the modes with $m \geq 2$. After the Fourier reduction step, the Ring Average technique is applied to $\tilde{\Phi}_{i,j}(k)$ to get the reconstructed magnetic flux $\tilde{\Phi}_{i,j}^r(k)$. The final magnetic flux is calculated as:

$$\Phi_{i,j}^r(k) = \tilde{\Phi}_{i,j}^r(k) + A_0 + A_1 \cos\left(\frac{2\pi}{N}k\right) + B_1 \sin\left(\frac{2\pi}{N}k\right). \quad (18)$$

3. Test results

In this section, we perform MHD simulations in cylindrical geometry with singularities to test the effectiveness of the Ring Average technique. The test simulations are done using the numerical kernel of the LFM code developed by Lyon et al. [6], which is a 3-D code solving the following ideal MHD equations on non-orthogonal curvilinear grids:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{u}) \quad (19)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} = -\nabla \cdot (\rho \mathbf{u} \mathbf{u} + \bar{\mathbf{I}} P) - \nabla \cdot \left(\bar{\mathbf{I}} \frac{B^2}{2} - \mathbf{B} \mathbf{B} \right) \quad (20)$$

$$\frac{\partial E_P}{\partial t} = -\nabla \cdot [\mathbf{u} (E_P + P)] - \mathbf{u} \cdot \nabla \cdot \left(\bar{\mathbf{I}} \frac{B^2}{2} - \mathbf{B} \mathbf{B} \right) \quad (21)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \cdot (\mathbf{u} \mathbf{B} - \mathbf{B} \mathbf{u}). \quad (22)$$

Dependent variables include: mass density (ρ), bulk velocity ($\mathbf{u}$), thermal pressure (P), magnetic field ($\mathbf{B}$), and plasma energy $E_P = \rho v^2/2 + P/(\gamma - 1)$ with $\gamma = 5/3$. Note that the LFM code solves the Cartesian component of the plasma velocity (u_x, u_y, u_z) such that the base directions of the velocity vector are grid-independent, while the normal directions of the face-centered magnetic fields are grid-dependent. In the Maxwell Solver, the LFM code solves the magnetic flux threading cell interfaces. Details about the numerical schemes implemented in the LFM code can be found in Lyon et al. [6]. The time stepping method used in the LFM scheme is the second-order Adams–Bashforth method, which combines a simple linear extrapolation step for predicting the middle time step and a full flux calculation for the corrector step. Therefore the Ring Average technique is only applied once after the corrector step. For other time stepping methods such as the TVD Runge–Kutta schemes, the Ring Average subroutines are needed at every stage of the time stepping in order to remove the large azimuthal wave number modes which violate the local CFL conditions. The effectiveness of the Ring Average technique is tested using two sets of test simulations: 2-D quasi-linear magnetic field-loop advection (field lines passing through singularity) and 2-D non-linear spherical blast wave (MHD shocks passing through singularity).

3.1. 2-D field-loop advection in cylindrical geometry

To test the effectiveness of the Ring Average technique dealing with quasi-linear advection of magnetic field lines across a cylindrical axis/pole, we use a magnetic field loop advection simulation in cylindrical geometry. The field loop advection simulation is very effective to show the capability of the Ring Average technique on preserving advecting magnetic field topologies. These test simulations also demonstrate the importance of keeping $\nabla \cdot \mathbf{B} = 0$ during the whole simulation, which are useful to show the effectiveness of the Ring Average technique on preserving $\nabla \cdot \mathbf{B} = 0$ both locally and globally. We initialize the advection of a circular field-loop by a constant velocity in a 2-D cylindrical domain. The simulation domain is defined as $r \times \theta = [0, 1] \times [0, 2\pi]$. The initial flow velocity is set to be 1.0, with $u_x = -\sin(\frac{\pi}{4})$ and $u_y = \cos(\frac{\pi}{4})$. The initial plasma density and pressure are both set to be 1.0, and the gas constant is $\gamma = 5/3$.

The magnetic field ($\mathbf{B} \nabla \times \mathbf{A}$) is initialized using a vector potential with $A_x = A_y = 0$, $A_z = \text{MAX}([A(R_0 - r)], 0)$ with its center located at $(r, \theta) = (0.4, -\pi/4)$. The amplitude of the field loop A is set to be 10^{-3} so that the magnetic pressure is much weaker compared to the plasma pressure, and the field loop is in (quasi) magnetostatic equilibrium. The radius of the field loop R_0 is set to be 0.3.

3.1.1. Field-loop through the singularity

First we focus on one single pass of a magnetic field-loop through the singularity by comparing the simulation results with and without the Ring Average technique. The cylindrical field loop advection simulations shown in the following section use 64×128 ($r \times \theta$) cells. Since it is a 2-D simulation with all the magnetic field components in the $x - y$ plane ($B_z \equiv 0$), the Ring Average process described in Section 2.2 is only needed for the B_θ component. The default number of chunks in each azimuthal ring near the pole is set to be [8 16 32 32 64 64 64 64]. This setup of the Ring Average means that 1) eight azimuthal rings near the pole need to apply the Ring Average procedure; and 2) in the inner most ring right next to the pole, there are 8 “thin” cells in each chunk, and the next one is 16, and so on. Note that the setup of averaging chunk for applying the Ring Average technique is not unique, the test simulation works reasonably with other setups such as [8 16 32 64] using four rings, resulting in a time step that is approximately 50% smaller than that using the eight-ring setup. The sensitivity of the field-loop advection to the choice of azimuthal chunks is presented in the next section.

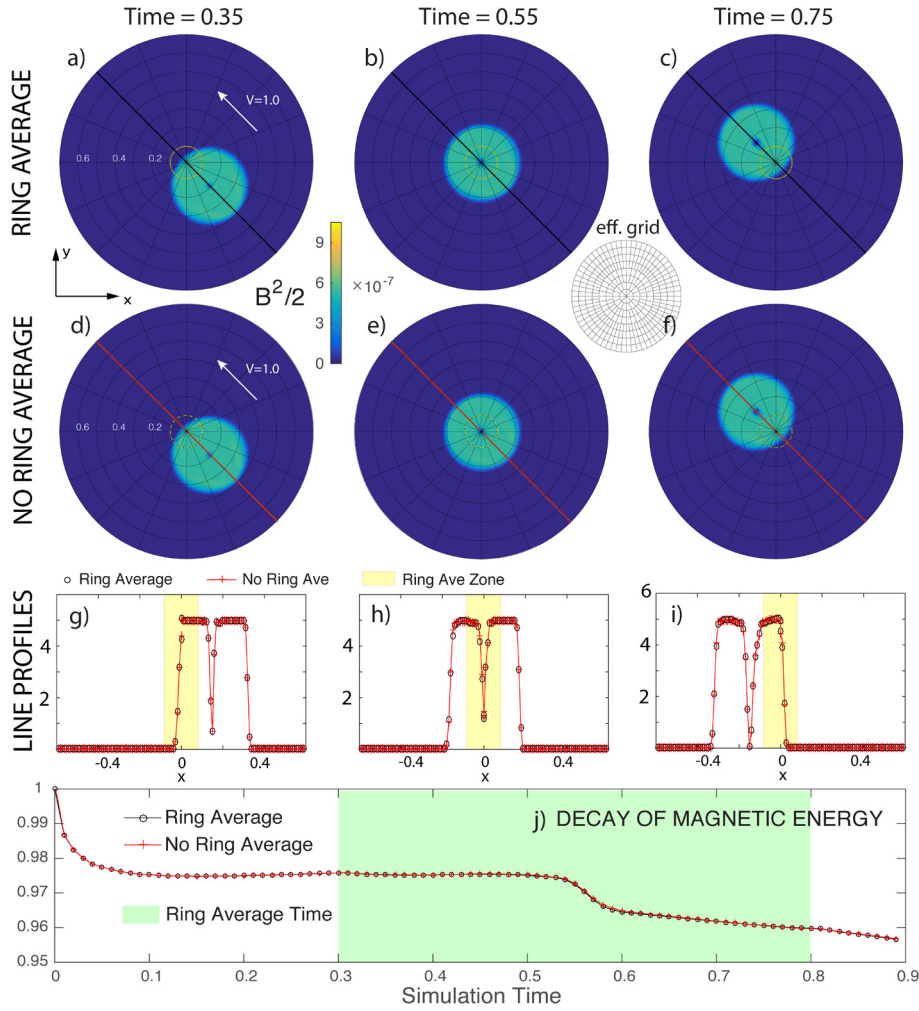


Fig. 6. The distribution of magnetic pressure $B^2/2$ at three simulation snapshots ($t = 0.35$, 0.55 and 0.75). Panel a)–c) shows the results using the Ring Average technique, and Panel d)–f) shows the corresponding results without the Ring Average technique applied. Panels g)–i) shows the comparison of distributions of magnetic energy along the line $y = -x$ at the three snapshots, respectively. Panel j) compares the decay of integrated magnetic energy with and without the Ring Average technique. The effect grid (distributions of azimuthal chunks) for the Ring Average technique is shown between Panel e) and f), with the number of averaging chunks in each azimuthal ring near the pole set to be [8 16 32 32 64 64 64].

Fig. 6a)–c) shows three snapshots of the magnetic field-loop at $t = 0.35$ (entering the Ring Average zone), $t = 0.55$ (centered at the Ring Average zone) and $t = 0.75$ (leaving the Ring Average zone) using the Ring Average technique. Unlike typical field-loop advection simulations in Cartesian geometries, the simulation time t indicates the “absolute” time during the advection of the field loop, rather than the time of periodic “grid crossings”. The color indicates the magnitude of the magnetic pressure $B^2/2$. The simulation results suggest that the Ring Average algorithm preserves the topology of the magnetic field loop quite well after passing through the pole. Fig. 6d)–f) shows the corresponding snapshots of the simulated magnetic pressure without applying the Ring Average technique. Compared with the non-Ring Average case shown in d)–f), the spatial distributions of magnetic pressure in a) to c) were not noticeably affected by the implementation of the Ring Average technique.

More quantitatively, comparisons of the profiles of magnetic pressure along the $y = -x$ line are shown in panels g)–i) and also confirm the effectiveness of the Ring Average technique. For this quasi-linear advection test problem, the local difference in the magnitude of magnetic energy along the $y = -x$ line was less than 5%, which occurred near the center of the Ring Average zone as indicated by the solid yellow circles in panel a)–c). On the other hand, the global difference, measured by the integrated magnetic energy as shown in panel j), was less than 0.1% during the simulations. In the end of the simulations, both integrated magnetic energy dropped to approximately 95.7% of the initial magnetic energy. Note that the simulation time during which the field loop encounters the Ring Average zone is shown using a shaded light green area in panel j), and a “maximum” difference ($< 0.1\%$) in the total magnetic energy occurred when the center of the field loop overlaps with the pole ($t = 0.55$). These quantitative comparisons on the total magnetic energy suggest that the Ring Average technique does not affect the solution of the magnetic field-loop advection problem significantly, especially when

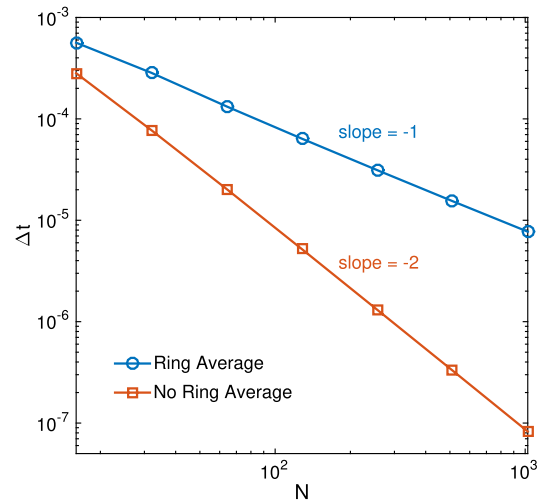


Fig. 7. The simulation time step Δt as a function of grid size $\frac{N}{2} \times N$ ($r \times \theta$).

Table 2

Ring Average setup used in the time step study. m is the index of rings in the radial direction, N_C is the number of chunks used in each azimuthal ring.

Index of rings	$m = 1$	2	3	4	5	6	7	8
$N = 16$	$N_C = 8$							
$N = 32$	$N_C = 8$	16						
$N = 64$	$N_C = 8$	16	32					
$N = 128$	$N_C = 8$	16	32	64				
$N = 256$	$N_C = 8$	16	32	64	128			
$N = 512$	$N_C = 8$	16	32	64	128	256		
$N = 1024$	$N_C = 8$	16	32	64	128	256	512	

a global measurement is used. Note that the Ring Average algorithm on the magnetic flux/field also keeps the total $\nabla \cdot \mathbf{B}$ at round-off level around 10^{-18} (not shown in Fig. 6), which is similar to the level of the simulated total $\nabla \cdot \mathbf{B}$ without the Ring Average technique.

Fig. 7 shows the relationship between the grid size N with $(\frac{N}{2} \times N$ cells) and the time step Δt in the magnetic field loop advection simulation (Δt does not change in the field-loop simulations), as the grid resolution increases from 8×16 to 512×1024 . The Ring Average setup for azimuthal chunks are listed in Table 2. When the Ring Average technique is switched on, the simulation time step drops linearly from 10^{-3} to 10^{-5} . While the Ring Average technique is switched off, the corresponding time step drops quadratically from 10^{-3} to 10^{-7} as expected. Therefore it is clear that the Ring Average technique relaxed the time step restriction in the azimuthal direction, which reduces the total amount of computations significantly, especially for 3-D simulations.

3.1.2. Sensitivity to the Ring Average configuration

Since the choice of the ring setup is not necessary unique, we performed another set of magnetic field-loop advection simulations to investigate the sensitivity of the solution to the configuration of the averaging chunks, which is also useful to test the robustness of the Ring Average technique.

Fig. 8 shows the distributions of magnetic pressure at $t = 1.0$ from five field-loop advection simulations using different choices of Ring Average configurations, together with a baseline case without ring average. Panels b) to f) show the simulation results with 3, 4, 6, 8, 11 averaging rings, respectively, while panel a) shows a reference simulation without Ring Average. The number of averaging chunks N_C within each azimuthal ring is listed in Table 3, and the corresponding effective grids based on the choice of azimuthal chunks are shown in the top left of each panel. In the simulation domain, the boundary of the Ring Average zone is indicated using solid red lines in each panel (except Panel a). It is clear that compared between the baseline simulation without Ring Average and the controlled simulations with relative aggressive Ring Average setups (8 and 11 rings), the simulated magnetic pressure are remarkably similar to each other in such color plots.

A more quantitative comparison between the six field-loop advection simulations is shown in Fig. 9 using the simulated profiles of magnetic pressure along a cut line $y = -x$ at $t = 1.0$. The spatial location of the line $y = -x$ for computing each profile is shown in the right corner of the plot. The baseline simulation is shown using a solid black line, and the colored circles show the profile from the ring average cases.

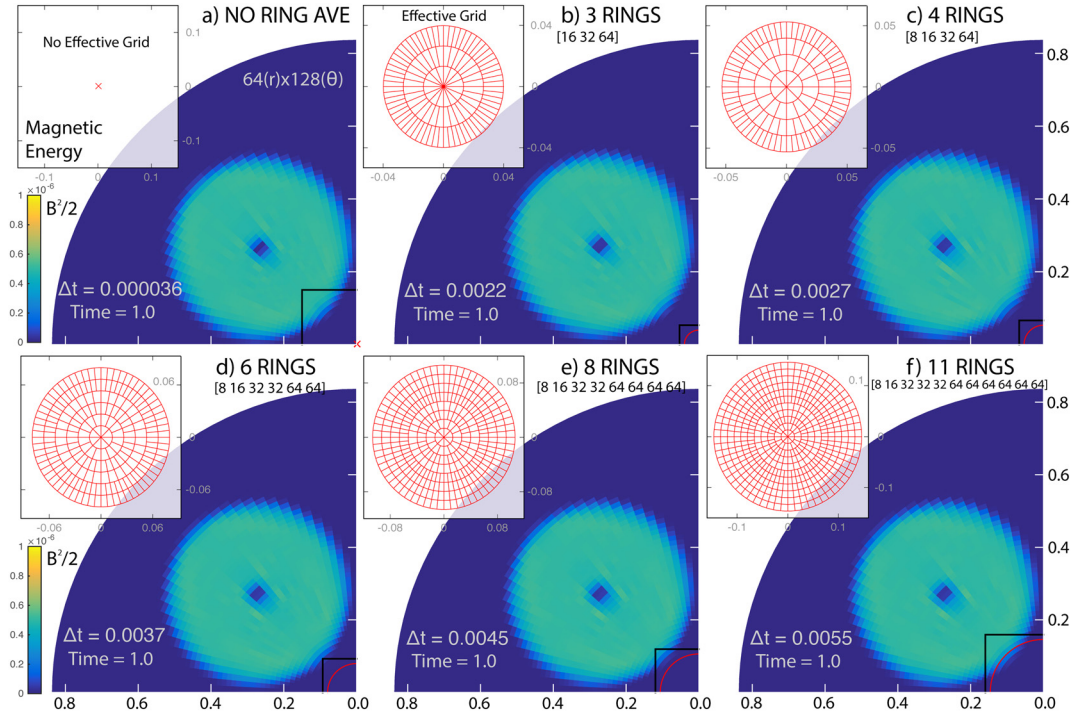


Fig. 8. The distribution of magnetic energy $B^2/2$ at simulation time $t = 1.0$ from simulations a) without the Ring Average technique and with Ring Average configurations using b) 3 rings, c) 4 rings, d) 6 rings, e) 8 rings and f) 11 rings. The effective grids based on the corresponding Ring Average configurations are shown in the top left of each panel, together with the number of averaging chunks within each ring listed in the top right of each panel. The simulation time step in each case is listed in the bottom left of each panel, and the boundary of the Ring Average zone is indicated using solid a red circle in the bottom left of each simulation domain.

Table 3

Ring Average setup used in the sensitivity study. m is the index of azimuthal rings in the radial direction, N_C is the number of averaging chunks used in each azimuthal ring.

Index of rings	$m = 1$	2	3	4	5	6	7	8	9	10	11
3 Rings	$N_C = 16$	32	64								
4 Rings	$N_C = 8$	16	32	64							
6 Rings	$N_C = 8$	16	32	32	64	64					
8 Rings	$N_C = 8$	16	32	32	64	64	64	64			
11 Rings	$N_C = 8$	16	32	32	32	64	64	64	64	64	64

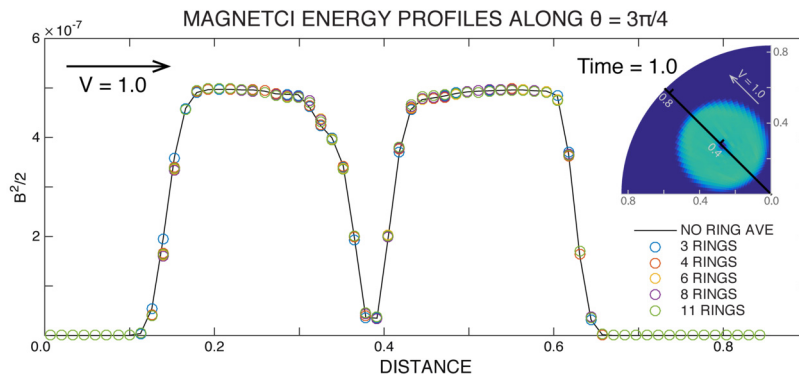


Fig. 9. The distribution of magnetic energy $B^2/2$ along the $y = -x$ line at $t = 1.0$ from simulations without the Ring Average technique (the black line) and with Ring Average configurations using 3 rings, 4 rings, 6 rings, 8 rings and 11 rings (the colored circles). The location of the cut line is shown in the top right of the figure.

These comparisons in Fig. 9 provide a quantitative measure on how much distortion was introduced due to the ring average method. As shown in Fig. 9, the most significant difference of magnetic energy along the $y = -x$ line occurs at the contact discontinuity just passed through the ring average zone (near $r = 0.15$). Local difference in magnetic energy using different Ring Average setups can be as large as 10% near the contact discontinuity. But again the evolution of the integrated magnetic energy was not significantly affected by the Ring Average configurations, as shown in panel j) of Fig. 6 (not shown here), which is a major advantage due to the conservation nature of the Ring Average technique. Note that since the simulation time step varies with ring average, it is difficult to take simulation snapshots at exactly $t = 1.0$ (the profiles shown in Fig. 9 were taken at the first simulation time step that is greater or equal to 1.0), e.g., the baseline case was taken at 1.000008 (27778 steps), while the case f) was taken at $T = 1.001$ (182 steps). This slightly mismatch in the actual simulation time for comparisons also contributes to the difference in the line profiles shown in Fig. 9, especially near the contact discontinuity where the profiles are steep. But overall, the quantitative comparisons in Fig. 9 suggest that the solution of the magnetic field-loop advection is not sensitive to the choice of the ring configurations. Although the Ring Average is quite robust, it is important to note that the appropriate implementation of the Ring Average configuration may be problem dependent, e.g., numerical experiments may be needed to find a balance between the accuracy of the solution and the additional computation introduced by the Ring Average technique in different research applications.

3.1.3. Multiple passes of a field-loop through singularity

In order to investigate accumulative effect of the field-loop advection simulation using the Ring Average technique (e.g. structures crossing a singularity multiple times) in long-term simulations, we adapted the field-loop advection simulation to a mapped cylindrical grid named the “squirrel” grid [8], and compared the solution with multiple crossings through a singularity with and without the Ring Average.

The setup of a “long-term” field-loop advection simulations in a logically cylindrical geometry is similar to those in the standard field-loop simulation as presented in Fig. 6. Instead of using a standard cylindrical simulation domain, the outer shells of the cylinder are continuously mapped from circles to square-like shapes, within the simulation domain defined as $|x| \leq 1$ and $|y| \leq 1$. Near the singularity centered at $(0, 0)$, the grid cells follow the cylindrical coordinates, and away from the pole the grids are gradually distorted to form a “square”-like domains in order to implement “periodic” boundary conditions to enable simulations with multiple crossings through the singularity (although the grid is not exactly periodic near the corners). Therefore the computational grid is highly non-uniform and non-orthogonal, which is handled by the high resolving power numeric schemes implemented in the LFM solver [6].

For the initial conditions, the center of the field loop was located at $(0, 0.4)$, with a radius of 0.3. The Amplitude of the magnetic field loop was set as $A = 10^{-3}$ to ensure quasi magnetostatic equilibrium. The initial density and pressure were set to be 1.0, and the initial velocity was in the y -direction with $v_y = -1$. The simulation grid consists 96 cells in the radial direction and 256 cells in the azimuthal direction. The number of averaging azimuthal rings was chosen as 5, with 8, 16, 32, 64 128 averaging chunks in each azimuthal ring starting from the innermost one near the pole. Periodic boundary conditions were implemented at $y = \pm 1$ where the top and bottom boundaries of the simulation domain were located.

Fig. 10 shows the simulated B_x component after crossing the singularity multiple times. Panels a)–d) show the spatial distributions of B_x at four snapshots right after passing through the pole ($t = 0.9, 2.9, 4.9, 6.9$). Panels e)–h) show the corresponding spatial distributions of B_x at the same simulation snapshots in a baseline simulation without applying the Ring Average technique. As shown by the color plots, although the distributions of B_x are distorted due to the irregularity of the mapped grid with increasing simulation time, the impact of the Ring Average technique on the overall solution is relative small. Panel i) shows the comparison on the decay of total magnetic energy between the two simulations with and without Ring Average. This quantitative comparison suggests that 1) the changes in the slope of the magnetic energy decay originated from either boundary or pole crossings, when the field loop encounters significant geometry changes; and 2) the difference on the total magnetic energy between the two simulations is relatively small, with peak difference lower than 0.1% during the pole crossing. Thus the accumulative error due to the ring average technique is negligible in this quasi-linear field loop advection.

However, it is very unlikely that accumulative errors in the field-loop test simulation presented in Fig. 10 can be conclusive for other long-term, non-linear MHD problems, although the Ring Average technique has been quite successful in climatological simulations of the Earth’s magnetosphere, which contains both shocks, smooth structures and their interactions within the simulation domain driven by upstream supersonic solar wind flows [9]. For long-term evolutions, necessary numerical experiments based on careful choice of the grid setups are needed when using the Ring Average technique in other non-linear MHD flow problems, e.g., the evolution of a magnetized accretion torus discussed in [10].

3.2. 2-D MHD blast wave in cylindrical geometry

To test the effectiveness of the Ring Average technique dealing with non-linear MHD shocks propagating through the pole, we use a standard MHD blast wave simulation in a strong initial magnetic field. The cylindrical simulation domain and grid resolution (64×128 cells) are the same as the magnetic field loop advection simulation discussed in the previous section, including the configuration of the Ring Average algorithm. The initial density is set to be 1.0, the initial plasma pressure is set to be 0.1, and the gas constant is $\gamma = 5/3$. Initial velocities are zero everywhere. Within the region $|\mathbf{r} - \mathbf{r}_0| < 0.1$, the plasma pressure is set to 10.0 (that is, 100 times the ambient pressure), where $\mathbf{r}_0 = (0.5, -\pi/4)$ in cylindrical

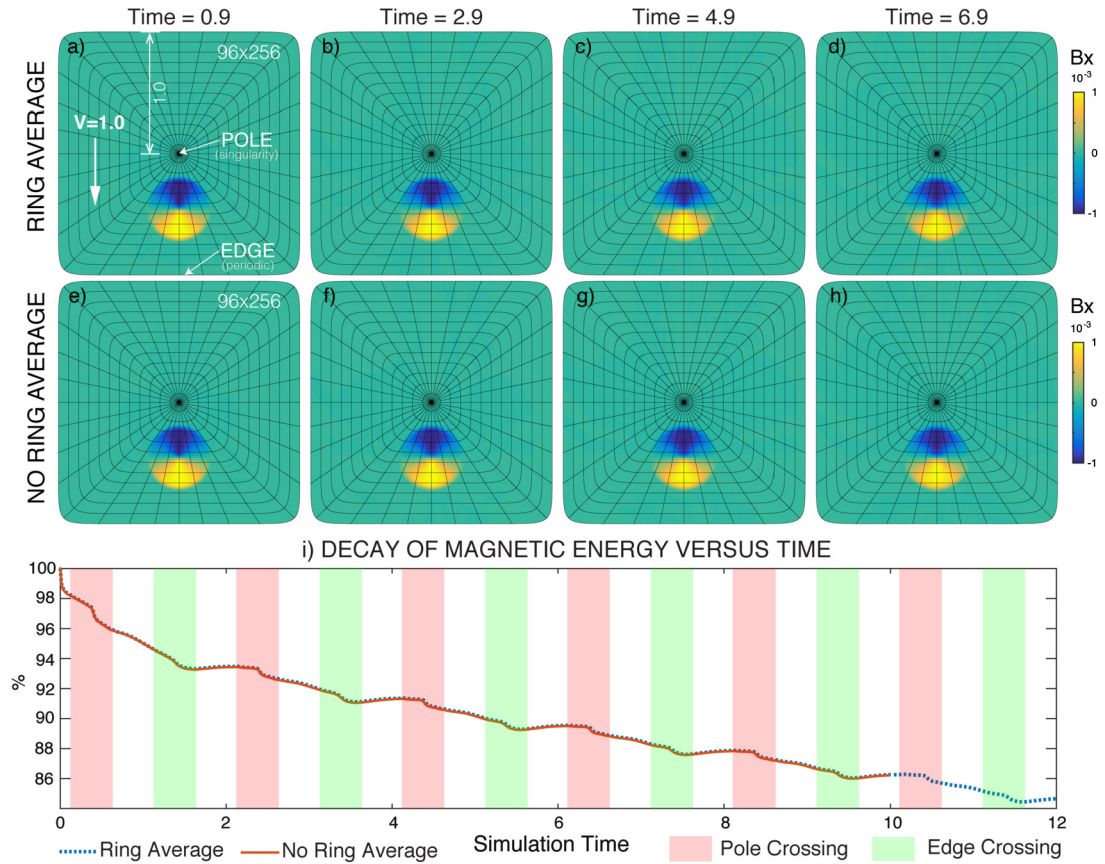


Fig. 10. The spatial distributions of the B_x component at simulation time $t = 0.9, 2.9, 4.9$ and 6.9 from simulations with Ring Average (panels a to d) and without Ring Average (panels e to h). Panel i) shows comparison on the decay of total magnetic energy with Ring Average (the blue dotted line) and without Ring Average (the red solid line). The pole crossing time periods are indicated using shaded red regions and the boundary crossing time periods are indicated using shaded green regions.

geometry. The initial magnetic field is chosen to be uniform in the simulation domain with $B_x = 1/\sqrt{2}$ and $B_y = 1/\sqrt{2}$, corresponding to an initial plasma $\beta = 0.4$. These initial conditions in the 2-D blast wave simulation launches MHD shocks propagating through the pole, thus the conservative nature of the ring average technique is crucial to obtain correct shock behavior near the pole.

3.2.1. MHD shocks pass through the singularity

In order to quantify the effectiveness of the Ring Average technique for non-linear MHD shock propagations. We performed two simulations of the spherical blast wave with and without the Ring Average technique, respectively. Fig. 11 shows the comparisons of the simulated magnetic energy between the two simulations. Panels a)–c) of Fig. 11 show the simulated spatial distributions of magnetic pressure at three different snapshots ($t = 0.02, t = 0.09$ and $t = 0.18$), respectively, using Ring Average. The configuration of the averaging chunks was the same as the setup used in the field loop advection test, which has eight rings with [8 16 32 32 64 64 64 64] averaging chunks from the first azimuthal ring to the eighth azimuthal ring, respectively. The boundary of the Ring Average zone is indicated using solid yellow circles in each panel. Panels d)–f) show the corresponding spatial distributions of magnetic energy at the same snapshots without Ring Average. Panels g)–i) in Fig. 11 shows the comparisons of the corresponding profiles of magnetic pressure along the cut line $y = -x$. The ring average zone is indicated in the panels g) to i) using shaded yellow regions.

Three representative snapshots are selected to show the behavior of MHD shocks before entering the Ring Average region ($t = 0.02$), passing the middle of the Ring Average region ($t = 0.09$) and after passing through the Ring Average region ($t = 0.18$). Compared with the non-Ring Average case shown in panels d)–f) of Fig. 11, the spatial distributions of the MHD shocks in panels a)–c) were not noticeably deformed by the implementation of the Ring Average technique, especially after passing through the Ring-Average zone. Note that in the MHD blast wave test simulation with constant initial magnetic fields, the Fourier reduction step before the averaging of magnetic fluxes described in Section 2.3 is very important to avoid possible artifacts originated from the clipping nature of the PPM reconstruction, which reduces errors in the evolution of magnetic flux near the singularity very effectively.

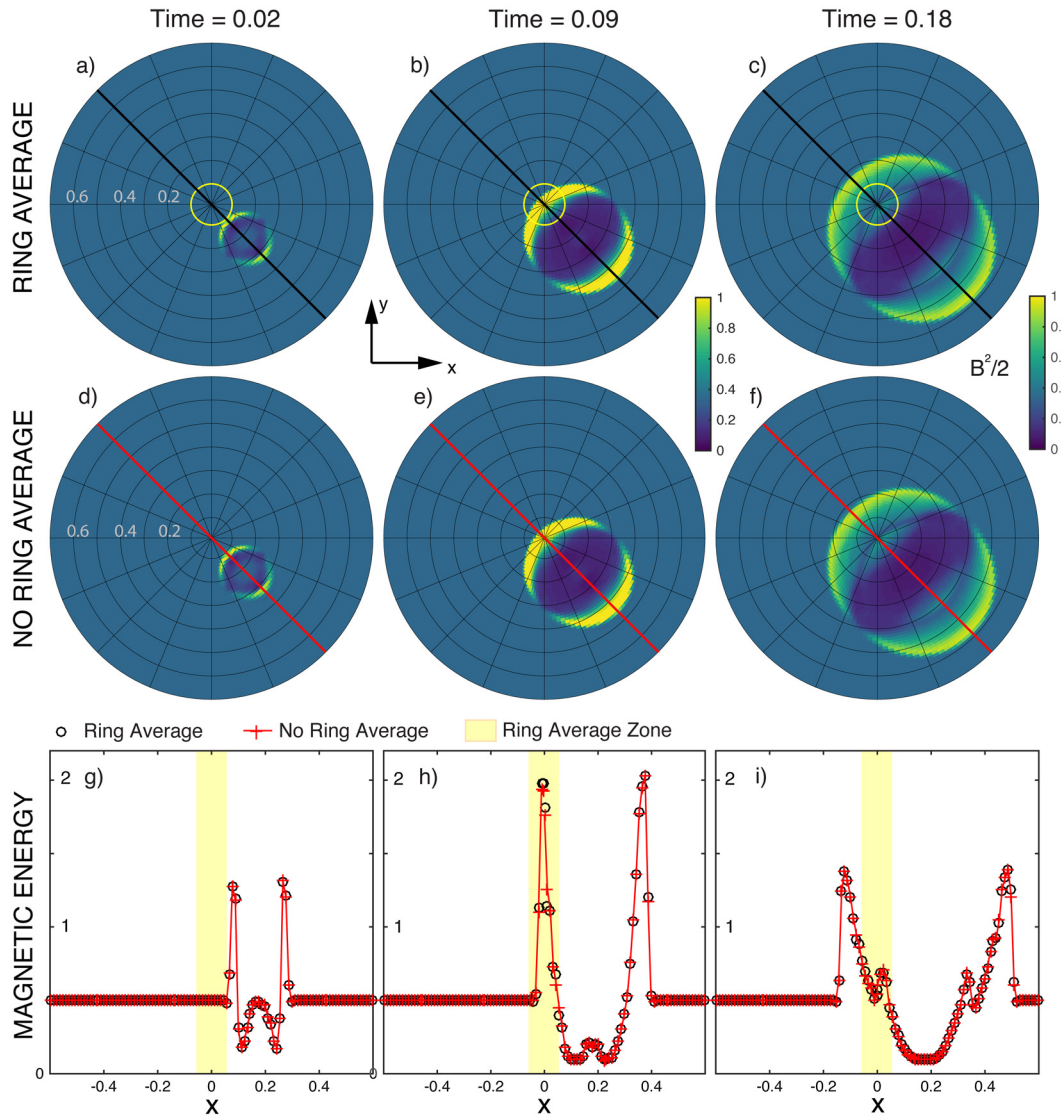


Fig. 11. The distribution of magnetic pressure $B^2/2$ at three simulation snapshots ($t = 0.02, 0.09$ and 0.18). Panels a)–c) show the results using the Ring Average technique, with the boundary of the corresponding Ring Average zones indicated using yellow circles. Panels d)–f) show the corresponding results without Ring Average. Panels g)–i) show the comparison of distributions of magnetic energy along the line $y = -x$ at the three snapshots, respectively. The number of averaging chunks in each azimuthal ring near the pole set to be [8 16 32 32 64 64 64].

Quantitatively comparisons on the line profiles shown in g) to i) indicate the effectiveness of the Ring Average technique. For this non-linear MHD blast wave problem, the local difference of magnetic energy within the ring average zone is greater than that in the field-loop advection problem, which was as high as 8% near the MHD shocks. This difference is possibly due to the azimuthal “smearing” of the magnetic flux during the redistribution step using the PPM algorithm. However, compared with the baseline simulation, the differences in the time evolution of total plasma and magnetic energy as shown in Fig. 12 were less than 0.1% during the whole simulation. Moreover, during the whole simulation, the maximum numerical variation in the total mass was as low as 10^{-10} , and the maximum numerical variations in the total momentum were around 10^{-8} , suggesting that the conservation nature of the MHD solver is preserved by the Ring Average technique. The conservation of energy is especially important for the behavior of MHD shocks passing through the singularity.

3.2.2. Sensitivity to the Ring Average configuration

A similar sensitivity study was done for the non-linear MHD blast wave simulations. The initial conditions were the same as the one analyzed in the previous section, and five different sets of chunk configurations were used to test the robustness of the ring average technique. The chunk configurations used in the simulations were the same as those shown in Fig. 8, with the outer boundary of the ring average zone indicated by a red solid line in each panel of Fig. 13.

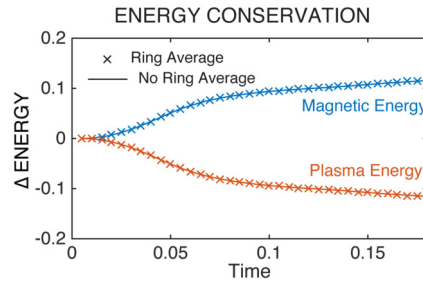


Fig. 12. The changes of total plasma (orange) and magnetic energy (blue) as functions of time in the MHD blast wave simulations. The baseline simulation results are shown using solid lines and the controlled simulation results with Ring Average are shown using crosses.

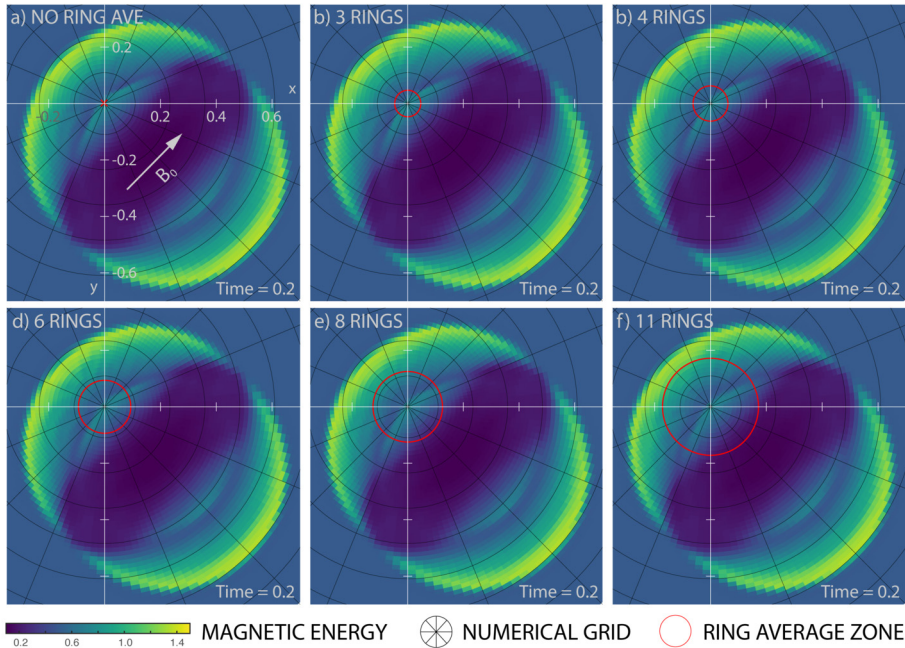


Fig. 13. The distribution of magnetic pressure $B^2/2$ at simulation time $t = 0.18$ from MHD blast wave simulations a) without the Ring Average technique and with Ring Average configurations using b) 3 rings, c) 4 rings, d) 6 rings, e) 8 rings and f) 11 rings. The effective grids based on the corresponding Ring Average configurations are shown in the top left of each panel, together with the number of averaging chunks within each ring listed in the top right of each panel. The simulation time step in each case is listed in the bottom left of each panel, and the boundary of the Ring Average zone is indicated using solid a red circle in the bottom left of each simulation domain.

Fig. 13 shows the spatial distributions of magnetic pressure derived from the MHD blast wave simulations at $t = 0.18$ using five different Ring Average configurations. The baseline simulation without the Ring Average technique is shown in panel a), while panels b) to f) show the corresponding simulation results using the Ring Average setups listed in Table 3, respectively (effective grids shown in Fig. 8). The simulation snapshots were taken at the time when a second MHD shock just reached the center of the ring average zone.

It is clear that in the non-linear blast wave simulations, regardless the choices of the ring average setup, MHD shocks passed through the singularity without significant distortions, which is primarily due to the conservation nature of the Ring Average technique. The simulated distributions of magnetic pressure at $t = 0.18$ were remarkably similar to the one without ring average. Note that at $t = 0.18$ when the second MHD shock passed right through the center of the ring average zone, it is slightly more “smeared” in the case with 11 rings compared to the baseline case, since the magnetic fluxes are redistributed in the azimuthal direction within a larger Ring Average zone, but the difference is relatively small which did not affect the global dynamics of the simulation such as the shape and the magnitude of the shocks

To compare the six simulations quantitatively, the profiles of the simulated magnetic pressure at $t = 0.18$ along the $y = -x$ line are shown in Fig. 14. The coordinate definition of the cut line is shown in the left corner of Fig. 14. The baseline simulation result is plotted using a solid black line, while the colored circles indicate the profiles from the five controlled simulations using different Ring Average configurations. These quantitative comparisons provide a measurement about how much distortion is introduced due to the ring average method for a highly non-linear MHD problems with shocks. Similar to the magnetic field-loop advection simulations, the most significant difference occurred at the front of the

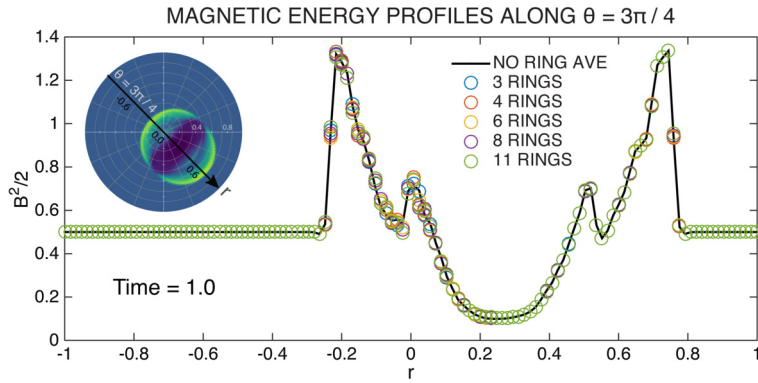


Fig. 14. The distribution of magnetic pressure $B^2/2$ along the $y = -x$ line at $t = 0.18$ from simulations without the Ring Average technique (the black line) and with Ring Average configurations using 3 rings, 4 rings, 6 rings, 8 rings and 11 rings (the colored circles). The coordinates of the cut line is defined in the top right of the figure.

MHD shocks moving crossing the Ring-Average zone. Maximum local difference in the magnitude of magnetic energy using different averaging chunk setups was around 5% near the shocks, but the changes in the integrated plasma and magnetic energy were not affected by the Ring Average setups (not shown here). Again the line profile comparisons suggest that the solution of the non-linear MHD blast wave problem is not significantly affected by the choice of the ring configurations, and the Ring Average technique is quite robust for non-linear MHD shock propagations. Note that since the Courant conditions were adjusted based on the effective grids, no spurious oscillations appear at the radial boundaries across Ring Averaging chunk domains in both the field-loop advection and blast wave simulations.

4. Discussions

The Ring Average technique, which can be highly non-linear, is roughly equivalent to applying an aggressive monotonic filter in the azimuthal direction at each ring in order to remove high azimuthal wavenumber oscillations from the solution. Therefore a larger time step can be applied based on the size of the averaging chunks rather than local “thin” cells. The filter form of the averaging step depends on the choice of the averaging chunks. The reconstruction step of the ring average technique is more complicated than just applying a filter since the PPM algorithm is non-linear and may create discontinuities as shown in Fig. 2. For example in special occasions such as step functions, the combined averaging and PPM reconstruction algorithms may reproduce the input step functions exactly without any filtering effect. Therefore, a rigorous derivation of the equivalent filter from the Ring Average algorithm is quite complicated. However, the following simplified analysis still provides some basic sense about the equivalent filter, based on the assumption that the input to the Ring Average algorithm contains only smooth structures (can we say that?).

In general, the ring averaging procedure has a couple of advantages. First, it can be applied as a post filter after a simulation step where the underlying code is unmodified. Second, a Fourier filter could be applied and global conservation would be preserved, but the conservation is non-local. The averaging scheme is non-local but only over the averaging zone. To maintain locality, the numerical fluxes could be filtered. This would require either making the fluxes available to the post-processing step, or doing the filtering in the main body of the code. A second, at least aesthetic, drawback is that simple digital filters lead to unsatisfying results. Starting with 1-D finite-volume discretization of the numerical solutions f_i , the change in conserved quantity within cell i can be written as

$$\Delta f_i = \frac{\Delta t}{\Delta x} (F_{i-\frac{1}{2}} - F_{i+\frac{1}{2}}), \quad (23)$$

where Δt is the time step, Δx is the average cell length, $F_{i-\frac{1}{2}}$ and $F_{i+\frac{1}{2}}$ are the interface fluxes at cell i ($i \in [0N - 1]$), respectively. If we “boxcar” average the solution f_i over L cells, the Courant condition will be approximately L times larger (assuming the CFL condition is dominated by the azimuthal direction), but the resulting change in Δf_i will be $\frac{\Delta t}{L\Delta x} (F_{i-\frac{1}{2}} - F_{i+\frac{1}{2}})$. This result means that the response in the center of the boxcar $L\Delta x$ comes only from the edges of the region, which may work for hydrodynamic simulations as illustrated by the “cell merging” method [4], but it does not seem optimal.

The averaging procedure within the chunks can be viewed as a filtering process in Fourier space. Assume that initial function can be represented by the nodal values at points, k , $f(\phi) = \sum_{k=0}^{N-1} f_k \delta(\phi - \phi_k)$, where $\phi_k = 2\pi k/N \equiv k\Delta$ is assumed to be the cyclic in the periodic azimuthal coordinate, N is the total number of azimuthal cells. The discrete Fourier transform of the $f(\phi)$ function is

$$\tilde{f}_m = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-imk\Delta} \quad (24)$$

The transform back to the configuration space is then written as

$$f(\phi) = \sum_{m=0}^{N-1} \tilde{f}_m e^{im\phi} \quad (25)$$

Using the PCM as an example, the averaging and reconstruction steps are combined in the Ring Average technique, which leads to the following piece-wise averaging function in the configuration space:

$$f_{\text{avg}}(\phi_{k'}) = \frac{1}{L} \sum_{j=1}^{L-1} f(\phi_{k'+j}), \quad (26)$$

where k' is the index of the center of the average area, L is the number of cells over which $f(\phi)$ is averaged. Assume M is the total number of averaging chunks, $M \times L = N$. In the configuration space, the averaged function is defined at M points ($k' \in [1, M]$) spaced by $2\pi L/N$ in the azimuthal coordinate ϕ . Thus to find the effective filter spectrum in the Fourier space, we back substitute the original configuration space function in term of the Fourier components. At each k' point, this substitution gives:

$$f_{\text{avg}}(\phi_{k'}) = \sum_{k=0}^{M-1} \frac{1}{L} \sum_{j=0}^{L-1} \tilde{f}_m e^{i(L \cdot k + j)\phi_{k'}}. \quad (27)$$

The sum over j can be evaluated to give

$$\tilde{f}_{\text{avg}}(\phi_{k'}) = \sum_{k=0}^{M-1} \tilde{f}_m e^{i(L \cdot k + j)\phi_{k'}} \frac{\sin(L\phi_{k'}/2)}{L \sin(\phi_{k'}/2)}. \quad (28)$$

In order to get the Fourier transform of the averaged function, we need to fill in the ϕ points around $\phi_{k'}$, this procedure gives

$$\tilde{f}_{\text{avg}}(m) = \frac{1}{N} \sum_{v=1}^{M-1} \sum_{\mu=1}^{L-1} f_{\text{avg}}(v\Delta) e^{i(L \cdot v + \mu)m\Delta} \quad (29)$$

Because of the orthogonality of the Fourier basis, the final result is simply

$$\tilde{f}_{\text{avg}}(m) = \tilde{f}_m \frac{1}{L} \left[\frac{\sin(Lm\Delta)}{\sin(m\Delta)} \right]^2 \quad (30)$$

The somewhat surprising result suggests that the Fourier amplitude for the averaging step is identical to a double boxcar convolution process in the configuration space. The difference is that the double boxcar has a phase function $e^{im(L-1)\Delta}$, that is missing from the piece-wise averaging step. The Fourier analysis could be extended to the reconstruction step but seems too complicated for the current purpose. Any of the averaged, or averaged and reconstructed, will also be subject to the non-linear TVD switches. Instead we show some simple examples of the effect of the reconstruction step on the amplitudes, without considering the effects of the non-linear TVD switchers near discontinuities. This is simply done by replacing the PCM in Equation (26) with either PLM and PPM and perform the same calculations from Equation (26) to (30). Fig. 15 shows the magnitude of the filter functions for $N = 128$ cells partitioned into $L = 8$ and $L = 32$ chunks using the PCM, PLM and PPM, respectively. The response is very sharply peaked for the low Fourier modes as expected. The very low frequency response is enhanced as the order of reconstruction is increased.

5. Conclusions

A high-order, conservative averaging-reconstruction method is developed to relax the severe time step restriction for explicit finite-volume MHD solvers on structured cylindrical/spherical mesh. The clustering of “thin” azimuthal cells problem is addressed through first averaging conservative quantities within these “thin cells” into azimuthal “chunks” and then reconstructing monotonically within each chunk, which gives much larger effective azimuthal length near the axis for the CFL condition. With the implementation of the ring average technique in finite-volume MHD solvers, the explicit time step Δt drops linearly with the grid resolution instead of quadratically, which reduces the total amount of computation significantly. The Ring Average technique can be implemented in curvilinear finite-volume MHD codes (either using staggered grids or not) as a post-processing step which requires no changes to the existing data structure, grid definition or numerical methods of the original solver. It is possible to adapt the Ring Average algorithm to conservative finite-difference MHD solvers. Also applications in other curvilinear geometry such as dipole geometry are also possible with appropriate modifications.

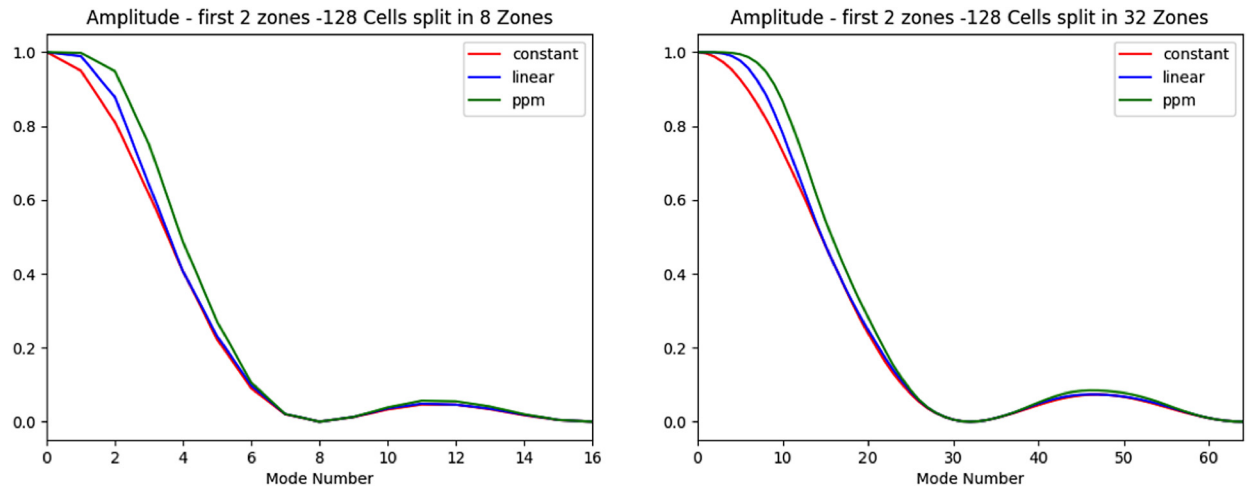


Fig. 15. The magnitude of the filter spectrum for $N = 128$ cells partitioned into left) $L = 8$ and right) $L = 32$ chunks using the PCM (red), PLM (blue) and PPM (green), respectively.

Acknowledgements

The research was supported by the National Center for Atmospheric Research (NCAR) under grant number NSFAGS-1404599 and the Applied Physics Laboratory at the Johns Hopkins University (JHUAPL). We would like to acknowledge high-performance computing support from Cheyenne ([doi:10.5065/D6RX99HX](https://doi.org/10.5065/D6RX99HX)) provided by NCAR's Computational and Information Systems Laboratory, sponsored by the National Science Foundation (NSF). Michael Wiltberger was serving at the NSF during the production of this paper. Any opinion, findings, or conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the NSF.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jcp.2018.08.020>.

References

- [1] K. Mohseni, T. Colonius, Numerical treatment of polar coordinate singularities, *J. Comput. Phys.* 157 (2000) 787–795.
- [2] S.H. Shih, D.R. Hixon, R.R. Mankbadi, Three dimensional structure in a supersonic jet: behavior near centerline, *Int. J. Comput. Fluid Dyn.* 13 (2000) 109–123.
- [3] R. Hixon, S.-H. Shih, R.R. Mankbadi, Numerical treatment of cylindrical coordinate centerline singularities, *Int. J. Comput. Fluid Dyn.* 15 (2001) 251–263.
- [4] R. Asaithambi, K. Mahesh, A note on a conservative finite volume approach to address numerical stiffness in polar meshes, *J. Comput. Phys.* 341 (2017) 377–385.
- [5] W.C. Skamarock, et al., A Description of the Advanced Research WRF Version 3, NCAR Tech. Note NCAR/TN-475+STR, 2008, 113.
- [6] J. Lyon, J. Fedder, C. Mobarry, The Lyon–Fedder–Mobarry (LFM) global MHD magnetospheric simulation code, *J. Atmos. Sol.-Terr. Phys.* 66 (2004) 1333–1350, Towards an Integrated Model of the Space Weather System.
- [7] P. Colella, P.R. Woodward, The piecewise parabolic method (PPM) for gas-dynamical simulations, *J. Comput. Phys.* 54 (1984) 174–201.
- [8] C. Fong, The conformal hyperbolic square and its ilk, in: Carlo H. Séquin, Douglas McKenna, Kristóf Fenyvesi, Eve Torrence, Bruce Torrence, R. Sarhangi (Eds.), *Proceedings of Bridges 2016: Mathematics, Music, Art, Architecture, Education, Culture*, Tessellations Publishing, Phoenix, Arizona, 2016, pp. 179–186, available online at <http://archive.bridgesmathart.org/2016/bridges2016-179.html>.
- [9] B. Zhang, W. Lotko, M. Wiltberger, O. Brambles, P. Damiano, A statistical study of magnetosphere–ionosphere coupling in the Lyon–Fedder–Mobarry global MHD model, *J. Atmos. Sol.-Terr. Phys.* 73 (2011) 686–702.
- [10] John F. Hawley, Global magnetohydrodynamical simulations of accretion tori, *Astrophys. J.* 528 (1) (2000) 462, IOP Publishing.